Chemistry Mini Project on Corrosion

E2 , E4 , E18 , E19 , E20 , E22

December 2025

1 Team Members

Roll.no	Name	Work Distribution	Signature
E18	Soham Gurav	Experimental Data Acquisition and Analysis	
E2	Aaryan Yadav	Report Lead: Writing and Formatting	
E4	Sharwari Bhagwat	Methodology and Results Documentation	
E20	Rugved Jagtap	Field Visit Coordination and Interview	
E19	Ishwari Bhawari	Theoretical Literature Review and Background Research	
E21	Vaibhav Jahane	Materials Management and Final Presentation Prep	

Faculty Name: Prof.Shreelekha Ghatte

This chemistry mini-project provided us with a comprehensive exploration of corrosion theory covering its fundamentals, various types, and essential prevention methods. We place a significant focus on electroplating, which is a key industrial technique to inhibit base metal degradation. Our study included both theoretical analysis and a practical educational field visit to a specialized electroplating facility to bridge academic knowledge with real-world application. To quantitatively assess corrosion rates, we created a data logging setup. Three iron nails were immersed in different environments : neutral(saltwater),acidic(acetic acid) and basic (saltwater with baking soda).We monitored the mass change, temperature, and visual characteristics over time. This experiment highlighted the dependence of the corrosion rate of a particular base metal on the pH of the surrounding medium. Furthermore, we investigated the real life implications of corrosion across several niche fields like rebar corrosion in bridges, accelerated automotive rust due to road salt, PCB failure in electronics and corrosion of orthopedic implants in biological systems. This broad approach underscores the widespread and costly nature of corrosion in engineering and science.

2 Introduction

2.1 Background

2.1.1 Thermodynamic Basis of Corrosion

Corrosion is fundamentally a thermo-driven process. According to the principles of thermodynamics, every system tends to move toward a state of lower Gibbs free energy. For most technologically important metals, this thermodynamically stable state corresponds to a **positive oxidation state** where the metal exists as an ion rather than a neutral atom.

2.1.2 Metals in Nature and Their Stability

In natural environments, metals are rarely found in their pure metallic state. Instead, they occur predominantly in *ores* as oxides, sulfides, or carbonates in which the metal atoms have already donated electrons and formed stable compounds. For example:

$$\mathrm{Fe^{2+}},\ \mathrm{Fe^{3+}} \longrightarrow \mathrm{Fe_2O_3},\ \mathrm{Fe_3O_4}$$

These oxidized forms represent the metal's low-energy, thermodynamically favored more stable states.

2.1.3 Refining and the Return to an Unstable State

During metallurgical extraction and refining, external energy must be supplied to convert metal ions back into neutral metal atoms. This is achieved through chemical reduction, electrolysis, or high-temperature processes that remove the associated non-metal species. The general reduction reaction may be written as:

$$\mathrm{M}^{n+} + ne^- \longrightarrow \mathrm{M(s)}$$

By accepting electrons, the metal returns to its *elemental* state, which carries a higher free energy compared to the corresponding oxide. Thus, refined metals are **thermodynamically unstable** with respect to their environment.

2.1.4 Corrosion as a Return to Stability

When the refined metal is exposed to the environment, it naturally tends to revert to its original low-energy state by releasing electrons. This oxidation process is represented by:

$$\mathrm{M(s)} \longrightarrow \mathrm{M}^{n+} + ne^-$$

The electrons released are consumed by a simultaneous reduction reaction in the environment, commonly involving oxygen and water:

$$\mathrm{O_2} + 2\mathrm{H_2O} + 4e^- \longrightarrow 4\mathrm{OH^-}$$

As a result, metal ions combine with oxygen and moisture to form corrosion products such as iron oxides (rust). The overall process leads to the gradual loss of metal from the surface and the formation of porous, brittle compounds that compromise mechanical integrity.

2.1.5 Corrosion as an Energy-Driven Phenomenon

Thus, corrosion can be understood as the metal's spontaneous tendency to transition from a high-energy (refined) state back to a thermodynamically favored oxidized state. The driving force for corrosion is the difference in Gibbs free energy between the metal and its corrosion products. From this standpoint, corrosion is not merely surface damage but a fundamental consequence of the metal's attempt to achieve stability through oxidation.

2.2 Importance of the Topic

The study of corrosion inhibition holds major significance, particularly in the context of defense engineering:

- **Exposure and Risk:** Critical defense assets, including rockets, missiles, naval vessels, and aircrafts, are constantly exposed to harsh environmental and operational conditions.
- **Impact on Readiness:** Any corrosion-induced degradation in these systems directly affects operational readiness, mission reliability, and personnel safety.
- **Vulnerability of High-Strength Alloys:**
  - Structures utilize lightweight, high-strength alloys (e.g., aluminum and titanium) for structural components and fuel tanks.
  - These alloys are susceptible to various aggressive forms of corrosion, including pitting, galvanic corrosion, and **stress corrosion cracking (SCC)**.
  - SCC is especially dangerous as it frequently originates in heat-affected zones, weld joints, or mechanically stressed regions.
- **Catastrophic Failure Risk:**
  - Even a minor reduction in the wall thickness of a propellant tank due to corrosion can become catastrophic.
  - During launch, high internal pressure and dynamic loads can cause sudden rupture and structural failure.
- **Essential Role of Inhibitors:**
  - Inhibitors play an essential role in stabilizing propellants and suppressing corrosive interactions between fuels/oxidizers and metallic containment materials.

– Effective inhibition ensures that ordnance remains structurally sound, chemically stable, and safe during long-term storage, thereby maintaining mission readiness and extending the service life of critical equipment.

2.3 Problem Statement

2.3.1 Electroplating (Electrodeposition)

To protect a metal surface from corrosion, various corrosion control techniques are employed. One of the most widely used methods is *electroplating* or *electrodeposition*. Electroplating is a process in which a coating metal (typically a more noble and corrosion-resistant metal) is deposited onto a base metal by passing a direct current (DC) through an electrolyte solution containing a soluble salt of the coating metal.

The noble metal coating protects the underlying base metal by acting as a barrier between the metal surface and the environment, thereby preventing direct contact with corrosive agents.

2.3.2 Principle of Electroplating

The process requires the base metal (to be protected) and the noble metal (to be deposited) to be immersed in the same electrolyte solution. The electrolyte must contain ions of the noble metal so that it can conduct electricity.

When an external DC power supply is connected to manipulate the flow of noble metal cations and its electrons towards the base metal:

- The **base metal** is connected to the **negative terminal** and acts as the **cathode**.
- The **noble metal** is connected to the **positive terminal** and acts as the **anode**.

At the anode (noble metal), oxidation occurs:

$$\mathrm{M}\ (\mathrm{s}) \rightarrow \mathrm{M}^{n+}(aq) + ne^-$$

The metal atoms of the noble metal lose electrons and enter the solution as cations.

These cations are attracted to the negative terminal and migrate through the electrolyte toward the cathode (base metal), which supplies electrons from the external circuit which were initially released by the noble metal and flow to the cathode by opposing the direction of current. At the cathode, reduction takes place:

$$\mathrm{M}^{n+}(aq) + ne^- \rightarrow \mathrm{M}\ (\mathrm{s})$$

Thus, the noble metal cations combine with electrons at the base metal surface and deposit as a uniform metallic layer.

2.3.3 Result

The overall effect is the gradual buildup of a coherent, adherent, and corrosion-resistant layer of the noble metal on the surface of the base metal. This deposited layer effectively isolates the base metal from environmental exposure, thereby preventing or significantly slowing corrosion.

3 Objectives

1. To understand the fundamental thermodynamic principles that govern the corrosion process.
2. To examine the importance of studying corrosion and its prevention, with emphasis on its impact on defense engineering materials, safety and long-term structural reliability.
3. To study the theoretical aspects of electrolysis and its application in corrosion prevention techniques such as electroplating and cathodic protection.
4. To explore the practical implementation of electrolysis-based corrosion prevention by conducting a field visit and observing industrial practices.
5. To gain hands-on experience by designing and assembling an experimental setup, performing measurements, logging data, and analyzing observations related to corrosion in different environments.

4 Literature Review

Corrosion of metals is a pervasive problem in engineering, affecting structural integrity, material lifetime and economic cost. A number of textbooks and review articles cover the mechanistic, thermodynamic and kinetic aspects of corrosion; this section summarises the findings of three key sources relevant to this mini-project: (1) the textbook by O.G. Palanna, (2) the article "Electrochemistry of Corrosion" from the Electrochemistry Encyclopedia, and (3) the web-resource "Physics Concepts – Corrosion Experiment" from the University of British Columbia.

4.1 Fundamental concepts (Palanna)

In his textbook, Palanna introduces corrosion as the chemical or electrochemical deterioration of metals when exposed to their environment. He defines corrosion as "the destruction or deterioration of metals by the surrounding through chemical or electrochemical changes".[1] Palanna distinguishes two broad types of corrosion:

- Dry (or chemical) corrosion: interaction of a metal with oxygen or other gases in a dry environment, leading to oxide formation.

- Wet (or electrochemical) corrosion: occurs in aqueous or humid environments, involves anodic and cathodic reactions, dissolution of metal ions and electron-transfer processes.

He then presents the electrochemical theory of corrosion in which a metal may serve as an anode (undergoing oxidation, e.g. $M \rightarrow M^{n+} + ne^-$) and concurrently a cathodic reaction occurs (such as reduction of oxygen or hydrogen evolution) in the environment. Furthermore, Palanna highlights the factors affecting corrosion rate: nature of the metal/alloy, nature of the corrosive medium (pH, conductivity, temperature), ratio of anodic to cathodic area, and the nature of corrosion products (whether protective film or not). He also covers methods of corrosion prevention: proper material selection, coatings (galvanisation, metal plating), cathodic protection (sacrificial anodes, impressed current) and inhibitors. In sum, Palanna provides a broad engineering-chemistry level overview of corrosion: mechanisms, types, influencing factors and control.

4.2 Electrochemical viewpoint (Electrochemistry Encyclopedia article)

The article by J. Kruger, "Electrochemistry of Corrosion", gives a detailed account of corrosion processes as fundamentally electrochemical in nature. He states that "corrosion is largely an electrochemical phenomenon" and that it involves electron transfer between a metal surface and an aqueous electrolyte.[2] Key points include:

- The driving force of corrosion is the tendency of metals to revert to lower energy ionic states, in contact with oxidising species (e.g., oxygen, water) in their environment.
- In aqueous solutions at ambient temperature, the metal surface often supports localized anodic and cathodic sites, and the flow of electrons and ions constitutes the corrosion cell.
- The article emphasises the consequences of corrosion: economic (large cost to national economies), safety (failure of structures, implants), technological (metallic components in devices)
- It also mentions the concept of passivity: some metals form protective oxide films (e.g., stainless steel, titanium) that slow or stop further corrosion. In short, this article gives a deeper electrochemical basis for corrosion — useful for your project if you are investigating metal dissolution, electrode potentials, cell formation, or protective film behaviour.

4.3 Demonstrative/experimental viewpoint (UBC "Physics Concepts" resource)

The webpage from the University of British Columbia outreach "Physics Concepts – Corrosion Experiment" provides a useful demonstration-oriented perspective of corrosion phenomena. [3] Salient aspects include:

- A set of metal samples (plain steel, copper, stainless steel, and steel with a zinc sacrificial anode) immersed in seawater or salt-water to observe differences in corrosion behaviour.
- The hypothesis that plain steel in seawater will corrode more quickly due to the conductive salt-water enhancing electron/ion flow; stainless steel will resist owing to chromium alloying forming a passive film; steel with a zinc anode will be protected since the zinc sacrifices itself (higher tendency to oxidize) thus preserving the steel.
- The practical set-up and visual demonstration emphasise the electrochemical cell concept (anode, cathode, electrolyte) in a tangible way. The write-up underscores that metal corrosion is fundamentally about the metal losing electrons to the environment.

This resource reinforces the mechanistic concepts from the two earlier sources, and is especially helpful if your mini-project includes an experimental component or demonstration.

4.4 Synthesis and gap identification

Putting together these sources gives a cohesive picture: corrosion is the process by which metals deteriorate via chemical and especially electrochemical reactions in their environments; the rate and mode depend on material, environment, protective films, and cell-like behaviour; control methods such as coatings and cathodic protection are widely used. Links between the sources: Palanna gives the broad engineering overview, Kruger delves into the electrochemical fundamentals, and the UBC demo provides an accessible illustration of how these ideas play out in practice. For your mini-project focus (on corrosion) you may note that:

- While Palanna covers many prevention methods, you may wish to deepen the electrochemical kinetics (current density, polarization behaviour, film impedance) which Kruger alludes to but does not fully cover.
- The demonstration resource shows simple experimental set-ups; you might consider a more rigorous measurement (e.g., rate of mass loss, electrode potential measurement, impedance) to align with the academic depth of Palanna and Kruger.
- None of the three sources focus deeply on certain advanced topics such as localised corrosion (e.g., pitting, crevice, stress-corrosion cracking) or modern electrochemical impedance spectroscopy methods; if your mini-project touches on these, you should supplement with peer-review articles.

In summary, these three sources provide a solid theoretical and demonstrative foundation for your corrosion mini-project; you may now build upon them with more specialised literature depending on your particular metal/alloy, environment, and protective method studied.

5 Methodology

In this experiment, three iron nails were suspended in three separate aqueous solutions representing acidic, neutral, and basic pH conditions. The objective was to investigate how variations in environmental pH influence the rate and extent of rust formation on iron.

5.1 Materials used

1. **Iron Nails (Three Units)**
   Commercially available iron nails of unknown surface coating were selected as the test specimens.
2. **Plastic Bottles (Three Units)**
   Transparent plastic bottles were used as individual reaction chambers corresponding to each pH condition.
3. **Cotton Threads (Three Units)**
   Threads were used to suspend each nail in its respective solution, ensuring no physical contact with the container walls or bottom.
4. **Neutral Medium: Standard Saltwater Solution**
   A neutral environment was prepared by dissolving **6 g NaCl (Salt)** in **220 mL** of distilled water, producing a **0.47 M NaCl solution** with an approximate **pH $\approx$ 7**.
5. **Acidic Medium: Dilute Acetic Acid Solution**
   The acidic environment consisted of **125 mL** of commercial vinegar containing **4.5% (w/v) acetic acid**, corresponding to an approximate concentration of **0.749 M CH_3COOH** and a measured **pH $\approx$ 2.5**.
6. **Basic Medium: Bicarbonate-Adjusted Saltwater Solution**
   The basic environment was prepared by adding **1.2 g $NaHCO_3$** to the standard saltwater solution, yielding a mildly alkaline medium with **pH $\approx$ 8.4**.

5.2 Experimental Setup

Each iron nail was suspended in its respective aqueous solution (acidic, neutral, and basic) using a cotton thread, ensuring that the nail remained fully immersed without touching the container walls or base. The three solution bottles were then placed in a cool, dry environment, away from direct sunlight or significant temperature fluctuations, and allowed to stand undisturbed for a period of one week to facilitate observable corrosion processes.

5.3 Procedure

After assembling the experimental setup, photographs of each bottle were taken at the start of the experiment to document the initial condition of the iron nails. The setup was then monitored daily for visible changes, and corresponding photographs were captured to track the progression of corrosion.

Ambient temperature in the experimental area was recorded each day to account for environmental variations. The pH of each solution was measured periodically to observe any drift from the initial values. At the end of the experiment, the nails were removed, cleaned, dried, and weighed to determine the mass loss corresponding to each pH condition. The solutions were tested for pH change by using a pH meter.

6 Observation and Results

6.1 Daily logging tables of setup

We labelled the three experimental bottles as **C1** (Neutral medium), **C2** (Acidic medium), and **C3** (Basic medium). The progression of corrosion in each medium was monitored through systematic daily logging. This included capturing photographs of the setup every day, recording the daily minimum and maximum ambient temperatures, and analysing the nail colour changes by processing the images using dedicated software. All observations were compiled into a comprehensive daily logging table to facilitate comparison across the three conditions.

(a) day1

(b) day2

(c) day3

(d) day4

(e) day6

(f) day9

(g) day11

(h) day14

(i) day15

6.2 Photos of Setup

Table 1: Daily Data Logging

Date	Day.No	Chamber ID	Temperature Range	Visual Rust Code	Observation Note
15.11.2025	1	C1	$Low = 12°C$ High=28°C	636864	water turns turbid
15.11.2025	1	C2		636864	clear water
15.11.2025	1	C3		636864	water turns turbid
16.11.2025	2	C1	$Low = 12°C$ High=28°C	5E5149	brownish water and screw thread forms visible rust
16.11.2025	2	C2		4B5053	clear water , no visible rust
16.11.2025	2	C3		433930	brownish water and screw thread forms visible rust
17.11.2025	3	C1	$Low = 11°C$ High=28°C	675446	
17.11.2025	3	C2		474644	
17.11.2025	3	C3		443A31	
18.11.2025	4	C1	$Low = 10°C$ High=27°C	49382E	Screw head forms visible rust layer
18.11.2025	4	C2		393B36	clear water , no visible rust
18.11.2025	4	C3		392C24	Screw head forms visible rust layer
19.11.2025	5	C1	$Low = 10°C$ High=29°C	685338	Water turns light brown
19.11.2025	5	C2		363229	clear water , no visible rust
19.11.2025	5	C3		3F2816	Water turns light brown
20.11.2025	6	C1	$Low = 12°C$ High=29°C	423222	Water gets more brownish
20.11.2025	6	C2		3A372E	clear water , no visible rust
20.11.2025	6	C3		312620	Water is light brown
21.11.2025	6	C1	$Low = 12°C$ High=30°C	62492B	Rust collected at bottom of bottle
21.11.2025	6	C2		332A23	clear water , no visible rust
21.11.2025	6	C3		382A21	Rust collected at bottom of bottle

6.3 Experimental Data

7 Discussion

7.1 Interpretation of Results

The visual inspection of the suspended iron nails revealed a counter-intuitive corrosion pattern across the three media. The nail immersed in the neutral medium (pH ≈ 7) exhibited the highest degree of visible rust formation, followed by the nail in the mildly basic medium. Surprisingly, the nail placed in the acidic solution showed the least apparent rusting, despite theoretical expectations that acidic environments typically accelerate corrosion.

This unexpected trend prompted a deeper investigation into the nature of corrosion products formed under varying pH conditions. While classical corrosion theory predicts rapid metal dissolution in acidic media due to enhanced hydrogen evolution and metal ion release, such environments may not always produce the characteristic reddish–brown hydrated iron(III) oxides commonly identified as "rust." Instead, acids often promote the formation of soluble iron salts that do not precipitate as visible corrosion deposits. Conversely, neutral and mildly basic conditions favor the precipitation of insoluble iron hydroxides and oxides, resulting in more observable rust, even if the actual rate of metal dissolution is lower.

Thus, our observations suggest that the discrepancy between expected corrosion rates and visible rust deposition arises from differences in the solubility and stability of corrosion products across pH levels, rather than a contradiction in corrosion kinetics. This realization motivated us to further analyze the specific corrosion products present in each medium and understand how the local pH environment influences not only the rate but also the *type* of corrosion occurring on iron.

7.2 Limitations and Challenges of Electroplating

1. Throwing Power and Uniformity

The precise nature of electrodeposition presents challenges in achieving reliable, uniform coating thickness:

- **Uneven Coating:** Electroplating tends to deposit the metal coating unevenly, especially on objects with complex geometries, sharp corners, or deep recesses.
- **Low Throwing Power:** The plating solution's throwing power its ability to deposit a uniform coating over the entire cathode surface is often poor. Areas closest to the anode receive a thicker, sometimes

excessively thick, deposit, while recessed areas (low-current density areas) receive a very thin or even no deposit at all. This compromises the corrosion protection in those critical recessed spots.

2. **Material Compatibility and Preparation**

   – **Substrate Restrictions:** Not all materials can be easily electroplated. For example, materials like aluminum or stainless steel require special, sometimes expensive and complex, pre-treatment steps (such as zincating for aluminum) to remove passive oxide layers before plating can adhere properly.

   – **Extensive Pre-treatment:** The process requires meticulous and time-consuming surface preparation (cleaning, degreasing, pickling, etching) to ensure good adhesion of the plated layer. Poor preparation leads to a weak, porous, or peeling coating.

3. **Environmental and Health Concerns**

   – **Toxic Chemicals:** Many common plating baths use toxic and hazardous chemicals, such as cyanides (for gold and copper plating) and chromium compounds (for hard chrome plating).

   – **Waste Disposal:** The resulting wastewater and sludge from the plating baths and rinsing stages must be treated to strict environmental standards, which adds significant cost and complexity to the operation.

4. **Microstructure and Defects**

   – **Porosity:** Electrodeposited coatings are often porous, especially when thin. These tiny pores can extend down to the substrate metal, creating pathways for corrosive media (like water and oxygen) to reach the base metal and start a localized corrosion process. This is a major limitation for corrosion resistance.

   – **Internal Stress:** The deposited layer can develop internal tensile or compressive stress, which can lead to cracking, warping, or peeling of the coating over time or under mechanical stress.

5. **Cost and Scale**

   – **Energy Intensive:** The process requires a constant electrical current for long periods, making it energy intensive and contributing significantly to operating costs.

- **Scale Limits:** While suitable for many sizes, electroplating can become challenging or uneconomical for extremely large structures or continuous, large volume applications that might be better served by alternative methods like hot dip galvanizing.

7.2.1 Scope for Improvement

Improvements in the Control of the Zinc Electroplating Process

1. Post-Treatment and Corrosion Resistance The quality of the final deposit is dictated by post-treatment, which is crucial for achieving requisite corrosion resistance:

- **Conversion Coating Transition:** Transition from toxic Hexavalent Chromium (Cr^{6+}) to environmentally compliant **Trivalent Chromium (Cr^{3+}) Passivation**. This provides comparable or superior corrosion performance while meeting current environmental standards.
- **Topcoat/Sealant Application:** Applying an organic or inorganic topcoat/sealer post-passivation significantly enhances corrosion protection, increasing resistance to harsh environments (e.g., road salt).

2. Plating Bath Formulation and Throwing Power

Improvements depend on the bath type, balancing efficiency and resistance:

- **Alkaline Non-Cyanide Zinc:** Due to inherently poor throwing power, focus on using specialized brighteners and levelers, and carefully adjusting anode placement to guarantee thickness uniformity across complex shapes.
- **Acid Zinc:** This bath offers faster plating but lower innate corrosion resistance. Improvement relies on strictly controlling bath additives (specifically chloride content) and prioritizing the quality of the subsequent passivation step.
- **Brightener Management:** Ensure consistent finish quality by installing dosing pumps or utilizing an ampere-hour (AH) meter to automatically replenish organic brighteners based on cumulative current passed.

3. Thermal and Mass Transfer Control

Precise physical control of the bath ensures consistent plating speed and crystal structure:

- **Consistent Agitation:** Utilize mechanical or air agitation to ensure fresh electrolyte reaches the cathode surface uniformly, preventing defects such as pitting and streaking.

- **Temperature Control:** Install reliable heating/cooling units to maintain the solution within its optimal processing range. This consistency is vital for maintaining the desired plating speed and deposit crystal structure.

4. Hydrogen Embrittlement Relief

For high-strength steel parts, hydrogen absorbed during the plating process poses a severe risk of delayed mechanical failure:

- **Mandatory Post-Bake:** Implement a strict post-bake (hydrogen embrittlement relief) step immediately following plating and before passivation. This critical safety measure involves baking the parts at temperatures between **180**°C and **220**°C for several hours to diffuse absorbed hydrogen out of the steel structure.

5.Improved Process Control (Related to Automation)

The quality of the electrodeposited metal coating is highly sensitive to fluctuations in operating conditions, necessitating precise control strategies.

- **Real-Time Monitoring & Automation:** This involves integrating modern sensors (such as pH probes, conductivity meters, and temperature sensors) with automated dosing systems. This shift replaces error prone manual checks, ensuring the bath composition and conditions are maintained within tight specifications (e.g., $\pm$ a few degrees or parts per million (ppm)), which leads to consistently higher coating quality.
- **Agitation Optimization:** This involves introducing advanced methods such as Ultrasonic Agitation or precise solution flow modeling to replace simple air sparging. Better agitation ensures the uniform mass transfer of metal ions across the cathode surface, directly improving the crucial characteristic of throwing power.

8 Conclusion

8.1 Key Findings

Explanation for the Absence of Visible Rust in the Acidic Medium

Although iron generally corrodes more rapidly in acidic environments, the nail placed in the acidic solution did not display the characteristic brown rust deposits. This behaviour can be explained by the chemical instability of iron corrosion products in low–pH conditions.

Rust primarily consists of hydrated iron(III) oxide, commonly represented as $Fe_2O_3 \cdot nH_2O$. This compound is unstable in acidic media. Acids contain a high concentration of hydrogen ions (H^+), which readily react with iron(III) oxides and convert them into soluble iron salts. The reaction can be represented as:

$$Fe_2O_3(s) + 6H^+(aq) \rightarrow 2Fe^{3+}(aq) + 3H_2O(l)$$

Thus, the following sequence occurs at the nail surface in the acidic medium:

1. Initial rust formation begins as iron oxidises.
2. The acidic solution immediately dissolves the newly formed rust, converting it into soluble Fe^{3+} ions.
3. These ions diffuse into the solution, preventing the accumulation of the solid brown iron oxide layer.

As a result, the nail *appears* not to have rusted, even though corrosion is actively occurring at the metal–solution interface.

Comparison Across Media

– **Acidic medium:** Dissolves rust as soon as it forms; no visible rust, but corrosion continues.

– **Neutral medium:** Provides ideal conditions for rust formation and deposition; thickest visible rust layer.

– **Basic medium:** OH^- ions promote formation of insoluble hydroxides and oxides; moderate visible rusting.

Real-Life Illustration

Acids such as vinegar or lemon juice are commonly used to remove rust from tools. Their acidic components react with iron oxides, dissolving the rust layer—often observable with mild effervescence. The behaviour observed in this experiment is consistent with this mechanism.

Conclusion

Acidic solutions do not inhibit corrosion; rather, they prevent the *visible* accumulation of rust by dissolving iron oxide layers as soon as they form. This explains the absence of brown rust on the nail in the acidic medium despite ongoing chemical attack.

8.2 Applications in Real World

8.2.1 Field Visit

Field Visit Description

As part of our project, we conducted a field visit to an industrial electroplating facility to gain first-hand insight into how corrosion-related problems are addressed in real-world applications. The visit enabled us to connect theoretical concepts with practical engineering practices.

During the visit, we examined the challenges posed by corrosion across the industry and observed how electroplating is employed as an effective method for corrosion prevention. We studied both the theoretical principles underlying electrolysis and their practical implementation on an industrial scale.

The technicians and engineers at the facility provided a detailed demonstration of the electroplating workflow, explaining each stage of the process, including surface preparation, cleaning and degreasing, electrolyte bath composition, electrode arrangement, controlled current application, and post-treatment steps. This step-by-step exposure allowed us to clearly understand how electrochemical deposition creates a protective metallic coating on components to enhance their durability and corrosion resistance.

Overall, the field visit greatly strengthened our understanding by bridging theory with practice and illustrating how scientific principles are transformed into reliable industrial techniques.

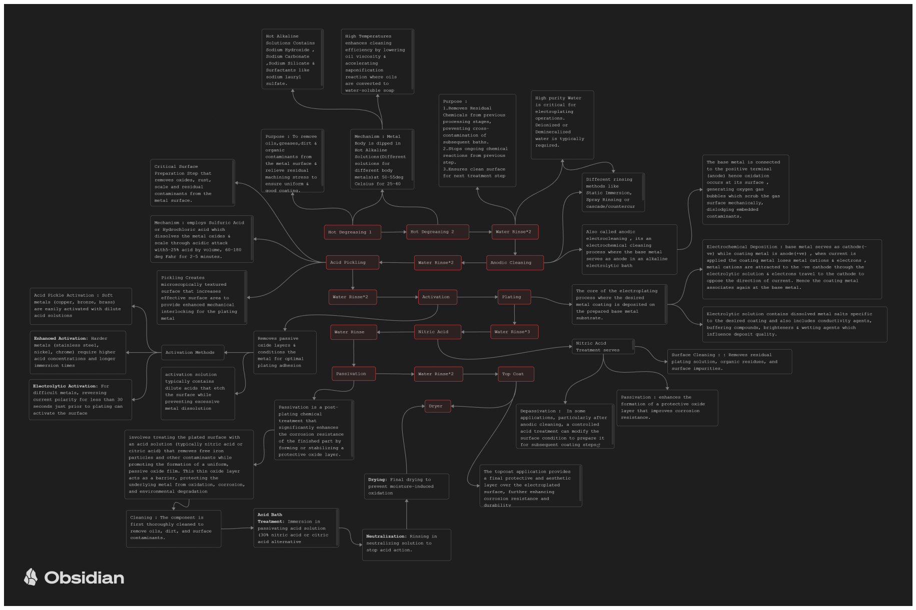
8.2.2 The Electroplating Workflow

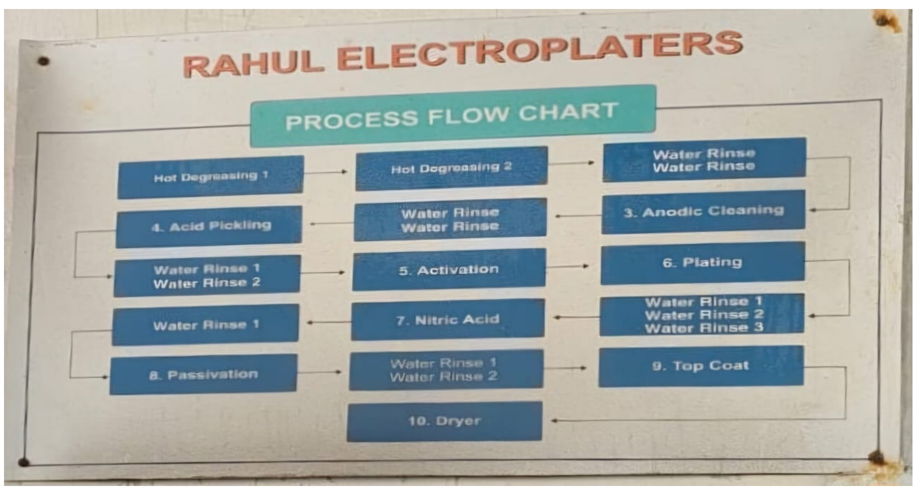
The theoretical description of electroplating found in textbooks typically presented as a straightforward, single-step deposition of a metal onto a substrate differs significantly from the procedures employed in industrial practice. In real manufacturing environments, electroplating is implemented as a highly structured, multi-stage workflow designed to ensure coating uniformity, adhesion strength, and long term durability. Each stage, from surface preparation and activation to controlled deposition and post treatment, plays a critical role in achieving a high quality protective or decorative finish. This contrast highlights the gap between simplified classroom models and the complex, quality-controlled processes required to meet industrial standards.

The process is as follows : Electroplating is a pivotal manufacturing process for depositing metal coatings with enhanced appearance and corrosion resistance. The steps are:

1. **Hot Degreasing (1 & 2):** Remove oils and greases via hot alkaline solutions (NaOH, surfactants). Dual stages ensure thorough cleaning.
2. **Water Rinse(s):** Wash away residues after each chemical process. Use deionized water to prevent contamination.
3. **Anodic Cleaning:** Electrochemical cleaning with workpiece as anode in alkaline bath. Oxygen evolution at the surface scrubs away embedded impurities.
4. **Acid Pickling:** Remove rust and oxides using sulfuric or hydrochloric acid. Mild etching increases adhesion.
5. **Activation:** Eliminate thin oxide layers and passivate surfaces using dilute acids or reverse-polarity electrolytic treatment.
6. **Plating:** Electrochemical deposition: Metal ions from electrolyte deposit onto cathode surface. Control current density, temperature, pH for optimal coverage.
7. **Nitric Acid Treatment:** Further cleans and, for some alloys, passivates by forming a protective oxide barrier.
8. **Passivation:** Acid treatment to promote a corrosion-resistant oxide film, especially for stainless steel and zinc.
9. **Top Coat:** Apply protective conversion coating (e.g., chromate), sealants, or paint for maximum durability and appearance.
10. **Drying:** Remove moisture with hot air or centrifugal dryers. Ensures defect-free finish and prevents flash rust.

Quality Control: Monitoring of chemical concentrations, rinse water purity, adhesion, and visual surface appearance at every stage is essential for reliable, corrosion-resistant plating.



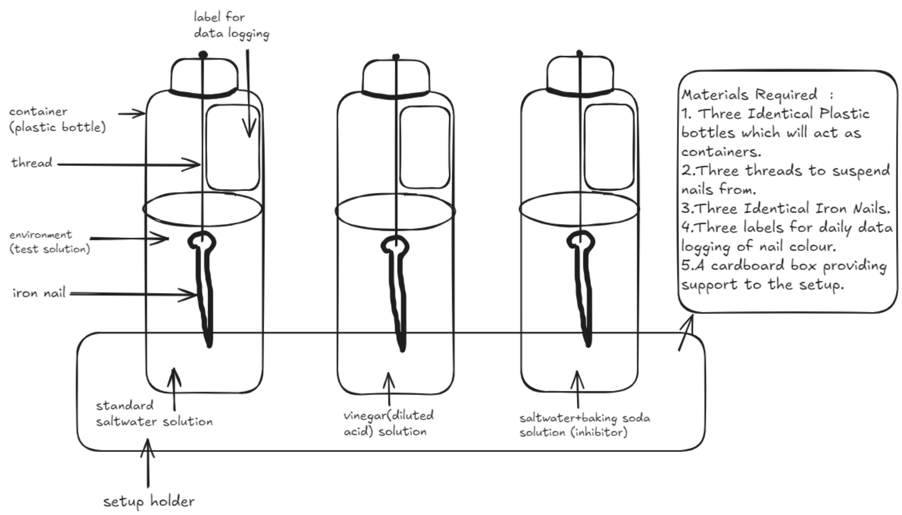


9 References

1.TextBooks referred 2.Research Papers referred 3.Websites

10 Annexures

10.1 Diagrams



10.2 Charts